Artem Semidetnov

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Education

MSc	**University of Geneva**, Mathematics	2025-2027 (expected)
BSc	**Saint Petersburg State University**, Mathematics	2021 – 2025
BSc	**Neapolis University Pafos**, Applied Computer Science, JetBrains Program	2022 – 2024

Research Interests

- Categorical homotopy theory; simplicial methods and operads; homotopy type theory and computer-formalized approaches; geometric and combinatorial group theory.

Publications and Preprints

An explicit non-elementary matrix over a bivariate Laurent polynomial ring Sept 2026
Vasily Ionin, ***Artem Semidetnov*** · arXiv:2609.04275

On Some More Problems from the Kourovka Notebook Aug 2026
Vasily Ionin, ***Artem Semidetnov*** · arXiv:2608.29219

The stable commutator length of a relator is not a one-relator group invariant Aug 2026
Artem Semidetnov · arXiv:2608.21465

The operad associated to a crossed simplicial group Dec 2025
Artem Semidetnov · arXiv:2512.14498

On the geometry of free nilpotent groups May 2021
Artem Semidetnov, Ruslan Magdiev · arXiv:2106.00095

AI-assisted research note
LLM tools were used in the research process for the three 2026 preprints listed above.

Research Experience / Projects

Operadic double dual
Preprint in preparation with Arshak Aivazian, Emmanuel Farjoun, and Vasily Ionin.

On Simplicial Resolutions for Link Groups Oct 2026
Independent mathematical note presenting two proofs of the Lei-Li-Wu theorem on a simplicial resolution of link groups.

JetBrains - Research Intern, HoTT and Dependent Types Lab Pafos, Cyprus
June 2024 - Sept 2024

- Developed the official Arend library and formalized results in algebra and homotopy type theory.
- Formalized the Eckmann-Hilton argument, Eilenberg-Mac Lane spaces, homology of types, group automorphisms, Schur's lemma, Maschke's lemma, and characterizations of group actions; selected contributions shipped in Arend 1.10.

Selected Talks

The operad associated to a crossed simplicial group Copenhagen, Denmark
July 2026

- Young Topologists Meeting 2026

Plus construction and simplicial methods — Pushkin, Russia, Aug 2025
- Conference "Algebraic Topology, Geometry, Combinatorics, and Data Analysis"

Twisting numbers on braid and Thompson groups — Nizhny Novgorod, Russia, 2024
- Topological Methods in Dynamics and Related Topics VII

On the Poisson boundary of lamplighter groups — Saint Petersburg, Russia
- St. Petersburg Seminar on Representation Theory and Dynamical Systems

On the geometry of free nilpotent groups — Novosibirsk, Russia, 2021
- Siberian Summer Conference

Honors and Awards

Excellence Master Fellowship, University of Geneva — 2025-2027

Rodnye Goroda Scholarship — 2023-2024

JetBrains Scholarship, Neapolis University Pafos — 2022-2024

Chebyshev Laboratory Travel Grant, Winter School in Mathematics and Theoretical Computer Science — 2025

Teaching Experience

Sirius Educational Center, Teaching Assistant, Groups of Intermediate Growth — Sochi, Russia, Apr 2024 – May 2024

Mathematics and Computer Science Faculty Program for Prospective Students, Teaching Assistant, Braid and Knot Theory — Saint Petersburg, Russia, June 2023

Laboratory for Continuous Mathematical Education, Mathematics Teacher for Gifted Students — Saint Petersburg, Russia, 2021 – 2023

Work Experience

JetBrains, Research Intern, HoTT and Dependent Types Lab — Remote, Nov 2025 – Apr 2026
- Designed and developed an MCP server for the Arend proof assistant, exposing type-checking and IDE functionality to AI coding agents.
- Integrated the MCP server with the Arend plugin for IntelliJ IDEA and built an experimental Arend coding agent using Koog.
- Fine-tuned models for Arend code completion.

IPONWEB (acquired by Criteo), Machine Learning Intern — Pafos, Cyprus, May 2023 – Sept 2023
- Reverse-engineered the behavior of a Criteo ML system that analyzes websites and produces word2vec representations; evaluated potential applications.

Workshops, Service and Additional Information

Teaching Higher Category Theory with Computers — ICERM, Brown University, Providence, RI, Aug 2026
Funded participant.

Academic Service and Outreach
- Organizer of the Euler International Mathematical Institute Functional Analysis Seminar.
- Invited judge, Saint Petersburg Tournament of Young Mathematicians (since 2021) and International Tournament of Young Mathematicians (2024, 2025).
- Problem author, ITYM 2024; Scientific Organizing Committee, ETEAM 2025.

Additional Information

- Finalist, Intel ISEF 2020; winner and PDMI special prize, Baltic SEF 2019; third place, International Tournament of Young Mathematicians 2019.
- English C1 (IELTS 8.0/9, taken in 2020 and 2024).